TECHNICAL REPORT

Emotional Response and Cognitive Load Analysis During MD02 Alarm Verification and Converter Panel Status Check

1. Objective of the Analysis

The objective of this analysis was to evaluate the emotional response: Stress, Cognitive Load, Trust and Engagement of four operators while performing the operational tasks:

1. Verification of the alarm "Mancanza Acqua Raffreddamento" on the MD02 panel
2. Operator checks the state of the "Start pompe Raffredamento" button on the Converter Panel.

The study aimed to identify:

- Assess emotional activation during operational alerts and verification tasks
- Identify the predominant emotion during the session for each operator
- Evaluate cognitive workload and uncertainty
- Examine situational confidence and perceived control
- Measure operator engagement
- Identify variability among operators
- Support human factors and operational safety improvements

This assessment contributes to a human factors evaluation intended to improve safety, operational efficiency, and system ergonomics.

2. Materials and Data Analyzed

Audio recordings captured during simulation sessions were analyzed.

Processed data:

- 4 operators voice recordings during training session simulations
- 4 Gamplays videos where the operators simulate the operational tasks through a VR application.

3. Method: BORIS and HUME.AI Implementation, to evaluate the timeline operators' emotions in two critical operational tasks and to analyze the behavior of Stress, Trust, Cognitive Load and Engagement.

3.1 Video Logs

BORIS

BORIS is an easy-to-use event logging software for video/audio coding and live observations. The following table shows the categories and description by operation tasks:

Table 1. BORIS setup

KEY	CODE	TYPE	DESCRIPTION	CATEGORY
l	LISTEN_ASSISTANT	Point event	Operator listen the TRAINING ASSISTANT	HUMAN-ASSISTANT INTERACTION
s	SPEAK_ASSISTANT	Point event	Operator speak the TRAINING ASSISTANT	HUMAN-ASSISTANT INTERACTION
h	HESITATION	State event	Operator hesitates WHAT TO DO	HESITATION
0	OPERATION_10	Point event	Operator checks if the allarm is OFF in the MD02 Panel	EXECUTION
1	OPERATION_1	Point event	Operator verifies the allarm on the MD02 Panel	EXECUTION
2	OPERATION_2	Point event	Operator checks the state of the "Start Pompe di Raffredamento button" on the Converter Panel	EXECUTION
3	OPERATION_3	Point event	Operator presses the "Apertura Interruttori" button in the Panel Converter	EXECUTION
4	OPERATION_4	Point event	Operator calls maintenance worker in the Control Room	EXECUTION
5	OPERATION_5	Point event	Operator presses the "Inserione Ausiliare" button in the Panel Converter	EXECUTION
6	OPERATION_6	Point event	Operator presses the "Chiusura Interruttore" button in the Panel Converter	EXECUTION
7	OPERATION_7	Point event	Operator turns the "Precarica Lever" in the Panel Converter	EXECUTION
8	OPERATION_8	Point event	Operator presses "Marcia Forno" button in the Panel Converter	EXECUTION
9	OPERATION_9	Point event	Operator presses the "Reset Allarmi" button in the Panel Converter	EXECUTION
e	PROCEDURAL_ERROR	State event	Operator procedural the TRAINING ERROR	ERROR
r	ASISTANT INTERACTION_ERROR	State event	Operator asistant interaction the TRAINING ERROR	ERROR

It was used to log the events throughout the gameplay in critical situations where operators have to interact with some scenario elements. It was obtained a task executing timeline with the logs of the critical events.

3.2 Voice processing

Voice recordings were processed using an emotional AI platform: Hume AI, which allows to evaluate paralinguistic vocal features and infer affective and cognitive states. Finally, It was obtained an excel file with the emotion's evaluation and the speech transcription timeline.

3.2 Variables Analyzed

The analysis was structured across four dimensions:

A. Stress

It has been confirmed that high levels of Distress and Anxiety lead to a drastic drop in operational performance. **(**Yerkes, R. M., & Dodson, J. D. (1908). The relation of strength of stimulus to rapidity of habit-formation)

Indicators:

- Anxiety
- Distress
- Fear
- Horror

These variables reflect emotional activation associated with alertness, operational pressure, and perceived risk.

B. Cognitive Load

Several papers on Multimodal Emotion Recognition demonstrate that vocal hesitations and confusion are correlated with increased heart rate and skin conductance during technical tasks. The use of emotions like Confusion and Doubt to measure the cognitive character is based on the work of John Sweller (Sweller, J. (1988). Cognitive load during problem solving: Effects on learning. *Cognitive science*, *12*(2), 257-285.)

Indicators:

- Concentration
- Confusion
- Doubt

These represent the mental effort required to process information and execute the task.

C. Trust

Trust in the virtual assistant is an affective state. If the user displays “Calm” while the assistant speaks, it means they trust the system. If they display “Distress”, they are "disillusioned" with the technology (Lee, J. D., & See, K. A. (2004). Trust in automation: Designing for appropriate reliance. *Human factors*, *46*(1), 50-80.)

Indicators:

- Calm
- Contentment
- Satisfaction

These relate to perceived control, confidence, and operational comfort.

D. Operational Engagement

Engagement is measured by the balance between challenge and skill. If hume has detected interest and excitement, the worker is in the so-called "state of flow."
According to Csikszentmihalyi, VR learning is only effective if the user is "engaged." Boredom or indifference indicates a workflow failure. (Csikszentmihalyi, M. (2011). Torrent flows the psychology of optimal experience definition. *Erfurt, Germany: University of Erfurt*.)

Indicators:

- Boredom
- Excitement
- Interest

3.3 Interpretation of Values

Hume outputs values on a normalized **0–1 scale**, representing the relative strength of vocal features associated with each state.

These values:

- do not represent absolute emotions
- are not psychological diagnoses
- are suitable for relative comparisons between operators and events

4. RESULTS

The simulation results indicate that the operators maintained effective operational performance across both tasks, with emotional and cognitive responses consistent with the demands and criticality of each scenario. The MD02 alarm verification elicited higher emotional activation and uncertainty, reflecting the urgency and perceived risk associated with alarm conditions, whereas the verification of the “Start Pompe di Raffreddamento” button on the Converter Panel generated lower emotional activation but required sustained attention and procedural confirmation. Across both tasks, operators demonstrated adequate engagement and maintained task focus, suggesting that the activities were perceived as operationally relevant. Variability was observed among individuals: one operator exhibited greater emotional reactivity and cognitive uncertainty, while another showed consistently stable responses and higher situational confidence, with the remaining operators displaying balanced profiles. These differences likely reflect variations in experience, familiarity with procedures, or interface interpretation. Importantly, no indicators of overload or disengagement were observed. The findings support the conclusion that the tasks impose functional levels of stress and cognitive demand, while also highlighting opportunities to improve alarm handling training, interface clarity, and procedural support to reduce variability and enhance operational reliability.